Xavier Lermusieaux

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Education

University of Ottawa Ottawa, Canada
BASc Software Engineering (COOP) *Expected: Dec. 2026*

- **Scholarships & Awards**: Dean's Honors List | Admission Academic Merit Scholarship
- **Relevant Courses**: DSA, Computer Architecture, Operating Systems, Software QA, Databases, Networking
- **Extracurriculars**: IEEE uOttawa VP Equity | Stratopore CAN-SBX Embedded | WaffleHacks '23 Winner

Experience

Ross Video Ottawa, ON
Software Developer - Robotics | React, TypeScript, Java Spring Boot *Term 1:Summer 2025 | Term 2: Summer 2026*

- Built an **interactive, canvas-based graphical editor** in **React + PixiJS** for designing physical equipment layouts, supporting **drag-to-move, resize, rotate, and vertex editing** with snap-to-grid, replacing a manual XML workflow and **cutting setup time by 6+ hours** per deployment.
- Implemented **optimistic UI updates** over a **WebSocket** protocol, applying and validating edits **client-side before server confirmation** with **automatic rollback** on failure, eliminating perceptible latency during **real-time editing**.
- Designed a **reusable 2D geometry and transformation system** (rotation, scaling, coordinate-space conversion) shared across all editor tools, **reducing duplicated logic** and accelerating delivery of new features.
- Optimized **rendering performance** for large interactive scenes via **bounding-box caching and viewport-aware scaling**, reducing redraw work and keeping the UI responsive under heavy load.
- Engineered **Spring Boot REST and WebSocket endpoints** backed by **MariaDB**, enabling reliable, low-latency data sync, and conducted **30+ peer code reviews** enforcing coding standards across the codebase.

Innovation, Science and Economic Development Canada (ISED) Ottawa, ON
Software and Operations Officer | Java, SQL, Apex, VBA, Salesforce *Term 1: Summer 2023 | Term 2: Fall 2024*

- Enhanced regression testing coverage to **85%** by designing and writing comprehensive **unit tests** for Salesforce, reducing manual QA effort by **40%**, and accelerating deployment cycles.
- Developed and optimized internal tools using **VBA and SQL**, automating repetitive tasks and **boosting team productivity by 90%** across multiple departments.
- Implemented a Salesforce emailing system notifying up to **5000 engineers** of KPR updates, **streamlining communication** across the organization.
- Collaborated with a 12-person engineering team to extend high-speed Internet to remote regions, reviewing LNDs and **improving connectivity for underserved communities**.
- Analyzed multi-million-dollar telecom invoices and verified program partner designs, optimizing **project delivery** and ensuring **on-time completion**.

Projects

BinGo | ***Go, WebSockets, Systems Programming, Ginkgo*** Winter 2026

- Built a **multi-platform visual concurrency debugger for Go**, helping developers inspect **goroutines, channels, mutexes, and synchronization behavior** through an interactive debugging workflow.
- Engineered the **end-to-end system architecture** in **Go**, including the core debugger engine, a **session-based WebSocket server/protocol**, and an interactive client for managing sessions, execution, and managing breakpoints.
- Implemented **cross-platform low-level debugging backends** for **Linux amd64 and macOS arm64**, using OS-specific process control and debug information to support **breakpoint handling** and runtime inspection.

QuickCV | ***Python, PyTorch, YOLOv8, OpenVINO, ONNX, OpenCV*** Winter 2026

- Trained and evaluated a YOLOv8 detector on a 35K-instance KITTI dataset, achieving **0.946 mAP@0.5** and **0.98 vehicle-class mAP**, with detailed precision/recall and confusion-matrix analysis.
- Designed a two-stage YOLOv8n→YOLOv8m cascade for CPU deployment, improving low-confidence detections by **+0.27 confidence** while avoiding second-stage inference for 70% of objects.
- Reduced inference latency by **47% (22.1s→11.6s)** and second-stage runtime by **57% (98ms→42ms)** through OpenVINO INT8 quantization, with **no measurable accuracy loss**.

Skills

Programming Languages: C++, C, Python, Java, SQL, TypeScript, Rust, Go
Spoken Languages: Native French and English
Development Tools : Git, Docker, Wireshark, Visual Studio, Android Studio, CMake, Gradle
Technologies: Firebase, Expo, Arduino, Linux, PostgreSQL, Unity
Frameworks/Libraries: React-Native, PyQt5, React.js, TensorFlow.js, Winsock, Spring Boot, Pytorch